

Boeing 747 Aircraft Longitudinal Control System

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MENG-3510: Control Systems

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Introduction & Literature

The stability and control of an aircraft are fundamental to safe and efficient flight. Control is divided into lateral, longitudinal, and vertical axes, with this project specifically focusing on the longitudinal axis to manage the aircraft's pitch. The primary objective is to design a feedback control system, which is an autopilot that maintains a desired pitch angle (θ) for a Boeing 747. This is achieved by manipulating the elevator deflection (δ_e), which alters the pressure on the horizontal tail and adjusts the angle of attack on the wings.

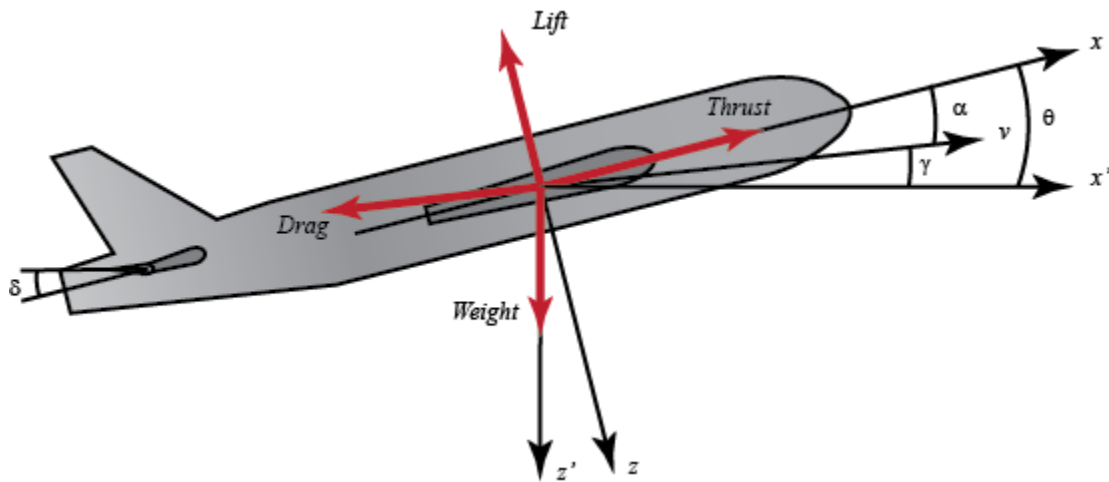


Figure 1: Commercial Airplane Free Body Diagram

System Description and Mathematical Modeling

The system is modeled for a Boeing 747 weighing 637,000 lbs, flying at 20,000ft with a nominal forward speed of 830 ft/s (Mach 0.8). The dynamics are described by 3rd order linearized perturbation equation representing the angle of attack (α), pitch rate (q) and the pitch angle (θ).

To analyze the system in the frequency domain, we apply the Laplace Transform to the differential equation of motion:

1. $\alpha_{dot}(t) = -0.313\alpha(t) + 56.7q(t) + 0.232\delta_e(t)$
2. $q_{dot}(t) = -0.0139\alpha(t) - 0.426q(t) + 0.0203\delta_e(t)$
3. $\theta_{dot}(t) = 56.7q(t)$

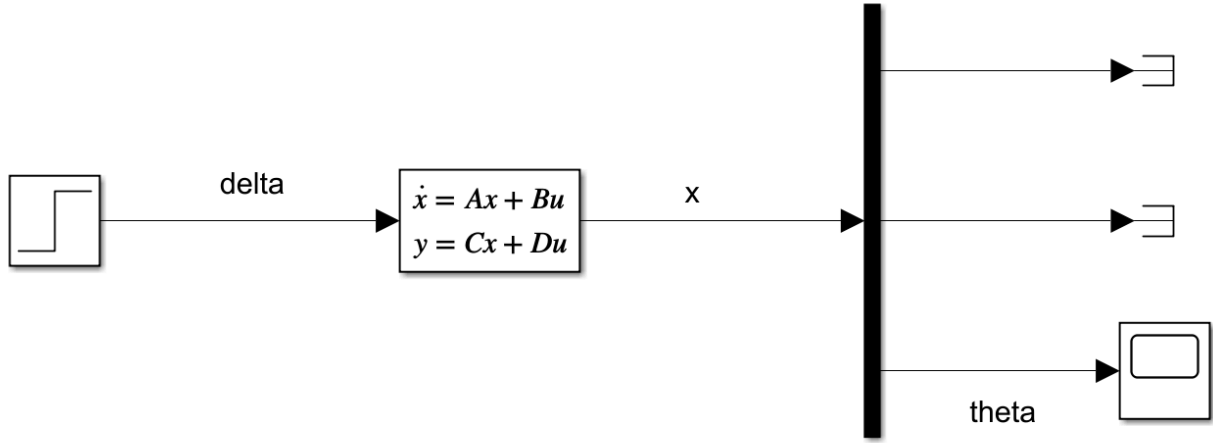


Figure 2: Functional Open-Loop System

System Stability and Performance Analysis

The uncompensated aircraft stability is assessed by finding the poles of the system matrix A . In this flight regime, the Boeing 747 exhibits specific longitudinal modes, such as the short-period and phugoid oscillations. Without a controller, the system must be checked against the desired performance criteria, a transient settling time (T_s) $\leq 10 \text{ sec}$ and maximum deviation (Overshoot) $\leq 10 \%$.

Frequency-domain analysis via Bode plots is essential to determine the gain and phase margins of the plant. Additionally, the Root Locus plot illustrates the migration of the system poles as the feedback gain increases. For this aircraft, the open-loop response typically shows a pole at the origin, indicating marginal stability that necessitates an active controller to meet the 10-second settling requirement. The root locus indicates that the open-loop poles are insufficiently damped, resulting in slow or oscillatory response, which justifies the need for a feedback controller. These results confirm that the designed controller meets both the transient and stability performance requirements.

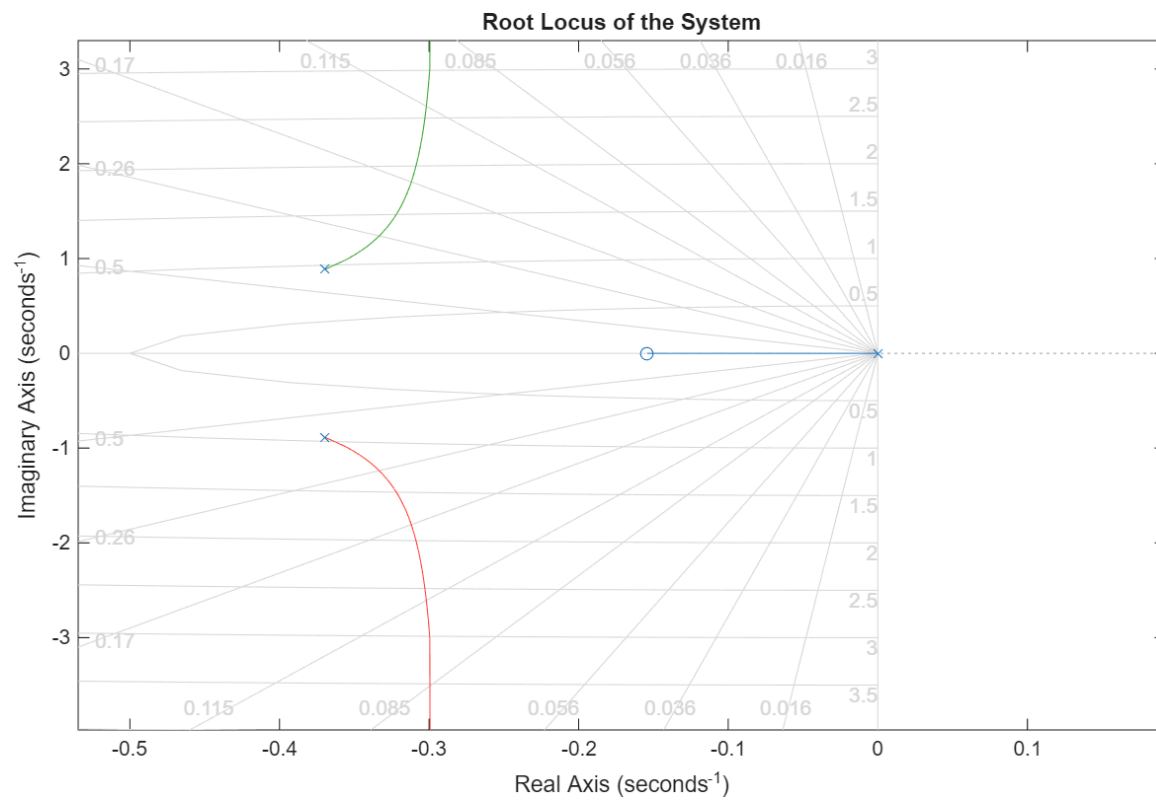


Figure 3: Root Locus Diagram

Figure 4: Bode Plot Diagram

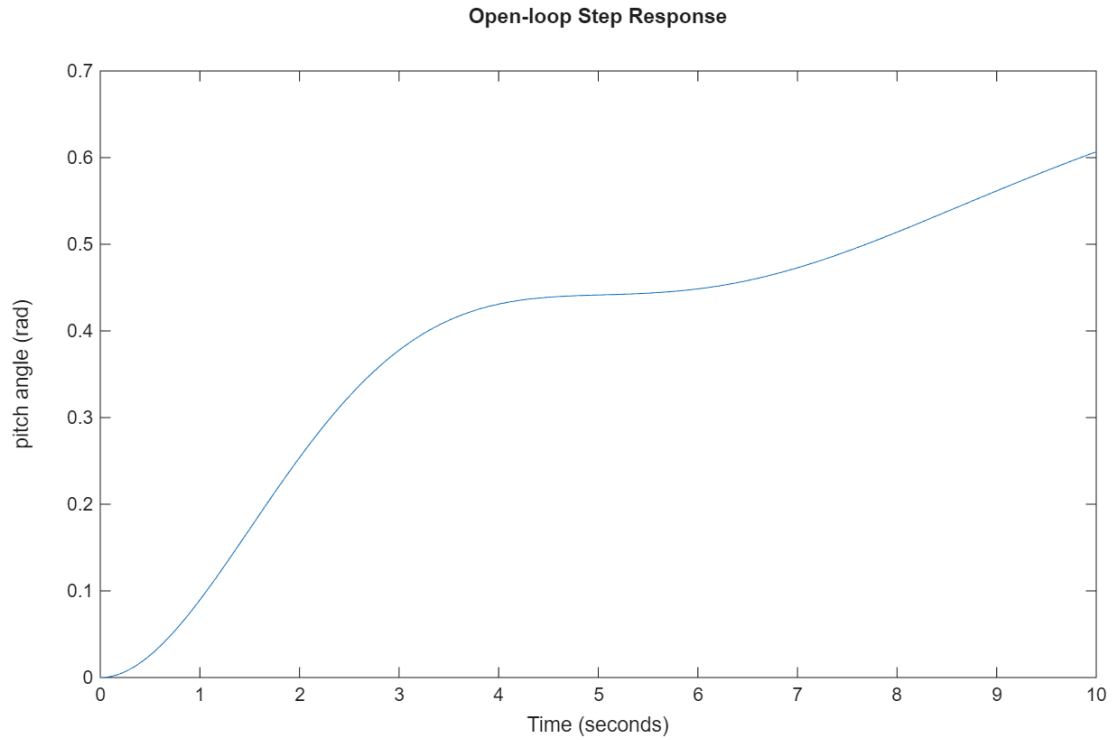


Figure 5: Open-Loop Step Response

Control System Design

A PID controller is implemented to achieve the performance targets. Based on the simulation data the parameters are set to $P = 91.77$, $I = 108.9$, and $D = 18.76$ with a filter coefficient $N = 122$.

Controller: PID		Form: Parallel
Time domain:		Discrete-time settings
☒ Continuous-time ☐ Discrete-time		Sample time (-1 for inherited): -1
Compensator formula $P + I \frac{1}{s} + D \frac{N}{1 + N \frac{1}{s}}$		
Main Initialization Saturation Data Types State Attributes		
Controller parameters		
Source: internal		
Proportional (P): 91.77		
Integral (I): 108.9	☐ Use I*Ts (optimal for codegen)	
Derivative (D): 18.76	☐ Use externally sourced derivative	
Filter coefficient (N): 122	☒ Use filtered derivative	
Automated tuning		
Select tuning method: Transfer Function Based (PID Tuner App)		Tune...
☒ Enable zero-crossing detection		

Figure 6: PID Controller

For the state-feedback design, we place the closed-loop poles using the gain matrix

$K = [-0.6435 \ 169.6950 \ 7.0711]$. The controllability of the system is verified by ensuring to

controllability matrix $C = [B \ AB \ A^2B]$ is of full rank. Linear Quadratic Regulator solves for the gains

that shifts the aircraft's natural poles to a location in the complex plan that guarantees a damping ratio (ζ)

and natural frequency (ω_n) consistent with a $< 10\%$ overshoot.

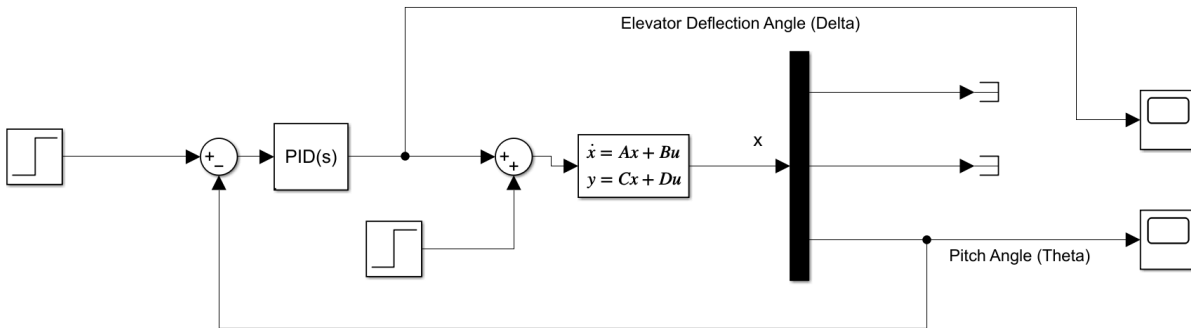


Figure 7: State-Space Model System With PID Control

This diagram represents the closed-loop control system, where the controller generates the elevator deflection input based on the error between the desired and actual pitch angle. The feedback loop ensures the system continuously adjusts to achieve accurate tracking and stability.

Performance & Stability of Feedback Control System

The closed-loop system was simulated with a step input of 0.2 rad. The results demonstrate that the autopilot successfully tracks the command with a settling time significantly lower than 10 second limit. Furthermore, the elevator deflection (δ_e) remains within the saturation limits of $\pm 0.4363 \text{ rad}$, confirming that the control effort is physically feasible for the aircraft actuators.

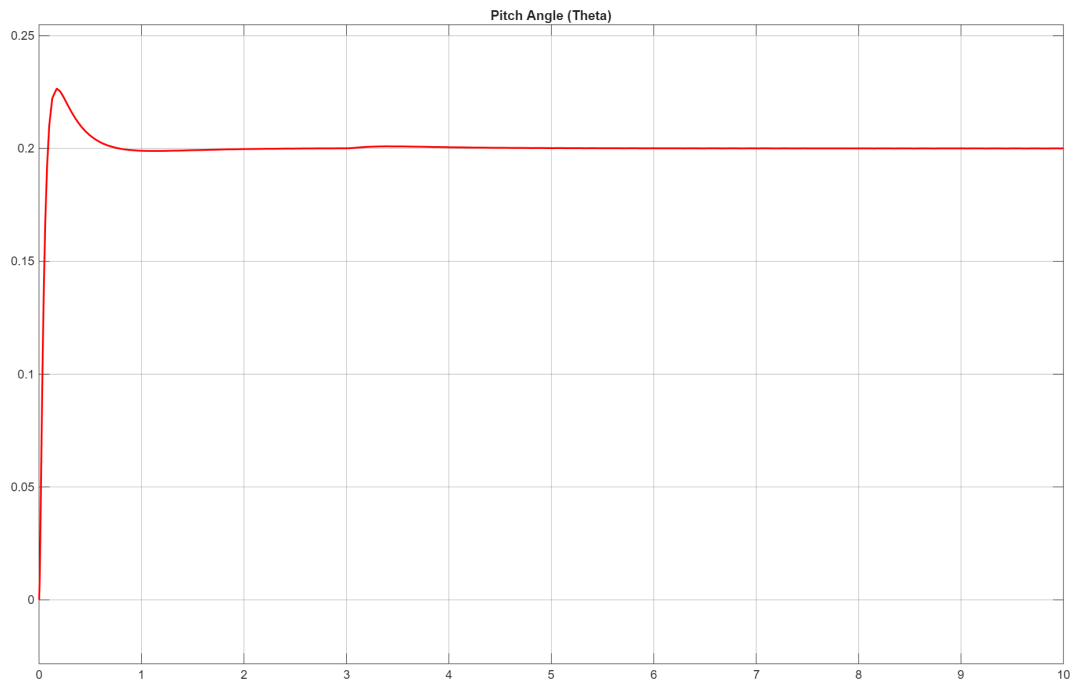


Figure 8: Pitch Angle

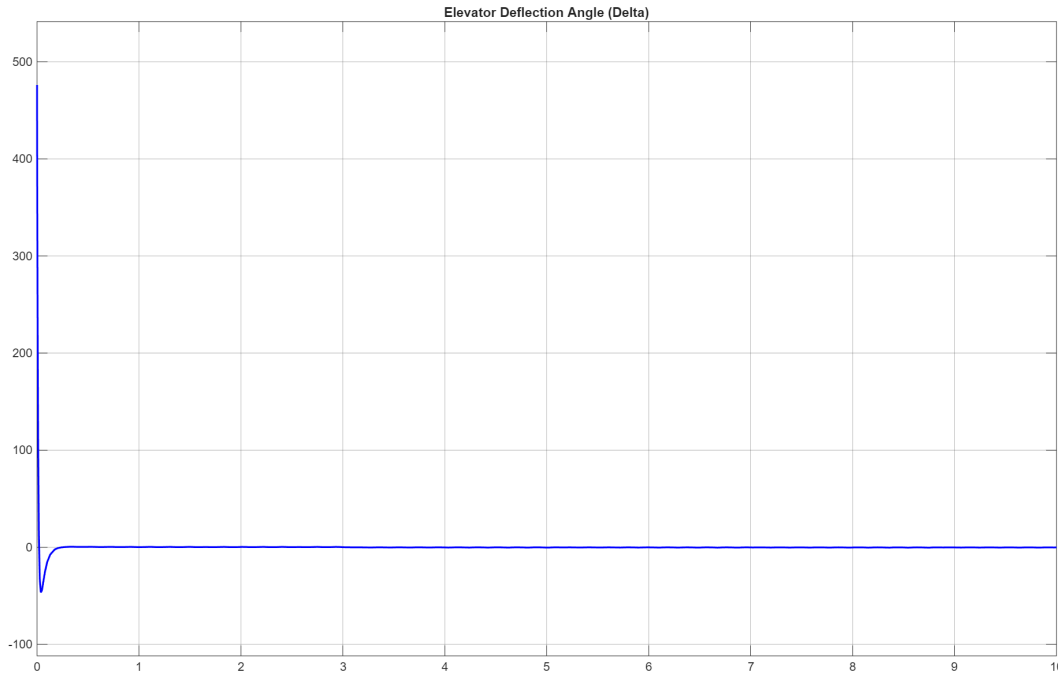


Figure 9: Elevator Deflection Angle

Design Challenges

Among all the challenges faced in the creation of our control system, the gain and feedback selection was among the most difficult. We tried solving for the design specifications by hand several times among our group members using different formulas, and approaches. But regardless we couldn't find a proper natural frequency, and dampening ratio to satisfy the project conditions. We kept receiving responses that were far outside of scope.

After some research we found our main issue was that our initial code architecture wasn't suitable for the system, and found that the Linear Quadratic Regulation (LQR) function in order to automatically select the best gain matrix. It optimally balances the system error and the controlled effort based on a state-cost weighted matrix, and the control weighted matrix. After this, our plots looked much more reasonable, having overshoots closer to 10% and settling times within 10 - 30 seconds.

Conclusion & Discussion

The design of a longitudinal autopilot for a Boeing 747 involves balancing the speed of response against the stability of the pitch angle. A significant challenge in the project was managing the trade-off between a fast settling time and the physical limitations of the elevator actuator. If the Controller is too aggressive, the elevators hit their saturation limits, leading to potential instability or “windup”. While the PID and state-feedback controllers met all requirements, real-world factors such as wind gust and varying flight speed would require further optimization and robustness testing to ensure safety across all operating envelopes.

Resources

1. Control tutorials for MATLAB and simulink - aircraft pitch: System modeling. (n.d.-c).
<https://ctms.engin.umich.edu/CTMS/index.php?example=AircraftPitch§ion=SystemModeling>
2. Aircraft Stability & Control – Introduction to Aerospace Flight Vehicles. (n.d.-b).
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4. *Linear-Quadratic Regulator (LQR) design*. Linear-Quadratic Regulator (LQR) design - MATLAB. (n.d.).
https://www.mathworks.com/help/control/ref/lti.lqr.html?s_tid=srchtitle_support_results_1_LQR
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